Proposition 1. *Given a concurrent monad T on an ordered category $\mathbb{C}$, $(\mathbf{Kl}(T), J)$, defined by*

- $|\mathbf{Kl}(T)| = |\mathbb{C}|$,
  $\mathbf{Kl}(T)(X,Y) = \mathbb{C}(X, TY)$,
- $k \leq^{\mathrm{K}} \ell$ *in* $\mathbf{Kl}(T)(X,Y)$ *iff* $k \leq \ell$ *in* $\mathbb{C}(X,TY)$,
- $\mathsf{id}^{\mathrm{K}}_X = \eta_X$,
  $k \mathbin{;^{\mathrm{K}}} \ell = k \,;\, T\ell \,;\, \mu_Z$ *for* $k \in \mathbf{Kl}(T)(X,Y)$, $\ell \in \mathbf{Kl}(T)(Y,Z)$,
- $\mathsf{jd} = \psi^0$,
  $k \parallel \ell = (k \otimes \ell) \,;\, \psi_{Y,V}$ *for* $k \in \mathbf{Kl}(T)(X,Y)$, $\ell \in \mathbf{Kl}(T)(U,V)$,
- $JX = X$
  $Jf = f; \eta_Y$ *for* $f \in \mathbb{C}(X,Y)$

is a concurrent category.

Moreover, K, defined by

- $KX = TX$,
  $Kk = Tk; \mu_Y$ *for* $k \in \mathbf{Kl}(T)(X,Y)$

is an ordered functor, monoidal-like in the appropriate sense, J is an ordered left adjoint to K, with unit the monad unit η and the counit $\varepsilon_X = \mathsf{id}_{TX}$, and $T = K \cdot J$.

Proof. $\mathbf{Kl}(T)$ is an ordered category:

$$k \leq \ell \quad \wedge \quad m \leq n \quad \Rightarrow \quad k \mathbin{;^{\mathrm{K}}} m = k \,;\, Tm \,;\, \mu_Z \leq \ell \,;\, Tn \,;\, \mu_Z = \ell \mathbin{;^{\mathrm{K}}} n$$

$$\begin{aligned} \mathsf{id}^{\mathrm{K}}_X \mathbin{;^{\mathrm{K}}} k &= \eta_X \,;\, Tk \,;\, \mu_Y \\ &= k \,;\, \eta_Y \,;\, \mu_Y \\ &= k \end{aligned}$$

$$\begin{aligned} k &= k \,;\, T\eta_Y \,;\, \mu_Y \\ &= k \mathbin{;^{\mathrm{K}}} \mathsf{id}^{\mathrm{K}}_Y \end{aligned}$$

$$\begin{aligned} (k \mathbin{;^{\mathrm{K}}} \ell) \mathbin{;^{\mathrm{K}}} m &= k \,;\, T\ell \,;\, \mu_Y \,;\, Tm \,;\, \mu_Z \\ &= k \,;\, T\ell \,;\, T(Tm) \,;\, \mu_{TZ} \,;\, \mu_Z \\ &= k \,;\, T\ell \,;\, T(Tm) \,;\, T\mu_Z \,;\, \mu_Z \\ &= k \,;\, T(\ell \,;\, Tm \,;\, \mu_Z) \,;\, \mu_Z \\ &= k \mathbin{;^{\mathrm{K}}} (\ell \mathbin{;^{\mathrm{K}}} m) \end{aligned}$$

As useful lemmata for proving the next points, we prove $Jf \mathbin{;^{\mathrm{K}}} k = f \,;\, k$ and $k \mathbin{;^{\mathrm{K}}} Jf = k \,;\, Tf$:

$$Jf \mathbin{;^{\mathrm{K}}} k = f \,;\, \eta_Y \,;\, Tk \,;\, \mu_Z = f \,;\, k \,;\, \eta_{TZ} \,;\, \mu_Z = f \,;\, k$$

$$k \mathbin{;^{\mathrm{K}}} Jf = k \,;\, T(f \,;\, \eta_Z) \,;\, \mu_Z = k \,;\, Tf \,;\, T\eta_Z \,;\, \mu_Z = k \,;\, Tf$$

$\mathbf{Kl}(T)$ is ordered monoidal-like:

$$k \leq \ell \quad \wedge \quad m \leq n \quad \Rightarrow \quad k \parallel m = (k \otimes m) \,;\, \psi_{Y,V} \leq (\ell \otimes n) \,;\, \psi_{Y,V} = \ell \parallel n$$

$$\begin{aligned} (\mathsf{jd} \parallel k) \mathbin{;^{\mathrm{K}}} J\lambda_Y &= (\mathsf{jd} \parallel k) \,;\, T\lambda_Y \\ &= (\psi^0 \otimes k) \,;\, \psi_{\mathsf{I},Y} \,;\, T\lambda_Y \\ &= (\mathsf{id}_{\mathsf{I}} \otimes k) \,;\, (\psi^0 \otimes \mathsf{id}_{TY}) \,;\, \psi_{\mathsf{I},Y} \,;\, T\lambda_Y \\ &= (\mathsf{id}_{\mathsf{I}} \otimes k) \,;\, \lambda_{TY} \\ &= \lambda_X \,;\, k \\ &= J\lambda_X \mathbin{;^{\mathrm{K}}} k \end{aligned}$$

$$
\begin{aligned}
k \mathbin{;^{\mathrm{K}}} J\rho_Y &= k \mathbin{;} T\rho_Y \\
&= k \mathbin{;} \rho_{TY} \mathbin{;} (\mathsf{id}_{TY} \otimes \psi^0) \mathbin{;} \psi_{Y,\mathsf{I}} \\
&= \rho_X \mathbin{;} (k \otimes \mathsf{id}_{\mathsf{I}}) \mathbin{;} (\mathsf{id}_{TY} \otimes \psi^0) \mathbin{;} \psi_{Y,\mathsf{I}} \\
&= \rho_X \mathbin{;} (k \otimes \psi^0) \mathbin{;} \psi_{Y,\mathsf{I}} \\
&= \rho_X \mathbin{;} (k \parallel \mathsf{jd}) \\
&= J\rho_X \mathbin{;^{\mathrm{K}}} (k \parallel \mathsf{jd})
\end{aligned}
$$

$$
\begin{aligned}
((k \parallel \ell) \parallel m)) \mathbin{;^{\mathrm{K}}} J\alpha_{U,V,W} &= ((k \parallel \ell) \parallel m)) \mathbin{;} T\alpha_{U,V,W} \\
&= (((k \otimes \ell) \mathbin{;} \psi_{U,V}) \otimes m) \mathbin{;} \psi_{U\otimes V,W} \mathbin{;} T\alpha_{U,V,V} \\
&= ((k \otimes \ell) \otimes m) \mathbin{;} (\psi_{U,V} \otimes \mathsf{id}_{TW}) \mathbin{;} \psi_{U\otimes V,W} \mathbin{;} T\alpha_{U,V,V} \\
&= ((k \otimes \ell) \otimes m) \mathbin{;} \alpha_{TU,TV,TW} \mathbin{;} (\mathsf{id}_{TU} \otimes \psi_{V,W}) \mathbin{;} \psi_{U,V\otimes W} \\
&= \alpha_{X,Y,Z} \mathbin{;} (k \otimes (\ell \otimes m)) \mathbin{;} (\mathsf{id}_{TU} \otimes \psi_{V,W}) \mathbin{;} \psi_{U,V\otimes W} \\
&= \alpha_{X,Y,Z} \mathbin{;} (k \otimes ((\ell \otimes m) \mathbin{;} \psi_{V,W})) \mathbin{;} \psi_{U,V\otimes W} \\
&= \alpha_{X,Y,Z} \mathbin{;} (k \parallel (\ell \parallel m)) \\
&= J\alpha_{X,Y,Z} \mathbin{;^{\mathrm{K}}} (k \parallel (\ell \parallel m))
\end{aligned}
$$

$$
\begin{aligned}
\mathsf{id}^{\mathrm{K}}_I &= \eta_I \\
&\leq \psi^0 \\
&= \mathsf{jd}
\end{aligned}
$$

$$
\begin{aligned}
\mathsf{jd} \mathbin{;^{\mathrm{K}}} \mathsf{jd} &= \psi^0 \mathbin{;} T\psi^0 \mathbin{;} \mu_{\mathsf{I}} \\
&\leq \psi^0 \\
&= \mathsf{jd}
\end{aligned}
$$

$$
\begin{aligned}
\mathsf{id}^{\mathrm{K}}_{X\otimes Y} &= \eta_{X\otimes Y} \\
&\leq (\eta_X \otimes \eta_Y) \mathbin{;} \psi_{X,Y} \\
&= \mathsf{id}^{\mathrm{K}}_X \parallel \mathsf{id}^{\mathrm{K}}_Y
\end{aligned}
$$

$$
\begin{aligned}
(k \parallel \ell) \mathbin{;^{\mathrm{K}}} (m \parallel n) &= (k \otimes \ell) \mathbin{;} \psi_{Y,V} \mathbin{;} T((m \otimes n) \mathbin{;} \psi_{Z,W}) \mathbin{;} \mu_{Z\otimes W} \\
&= (k \otimes \ell) \mathbin{;} \psi_{Y,V} \mathbin{;} T(m \otimes n) \mathbin{;} T\psi_{Z,W} \mathbin{;} \mu_{Z\otimes W} \\
&= (k \otimes \ell) \mathbin{;} (Tm \otimes Tn) \mathbin{;} \psi_{TZ,TW} \mathbin{;} T\psi_{Z,W} \mathbin{;} \mu_{Z\otimes W} \\
&\leq (k \otimes \ell) \mathbin{;} (Tm \otimes Tn) \mathbin{;} (\mu_Z \otimes \mu_W) \mathbin{;} \psi_{Z,W} \\
&= ((k \mathbin{;} Tm \mathbin{;} \mu_Z) \otimes (\ell \mathbin{;} Tn \mathbin{;} \mu_W)) \mathbin{;} \psi_{Z,W} \\
&= (k \mathbin{;^{\mathrm{K}}} m) \parallel (\ell \mathbin{;^{\mathrm{K}}} n)
\end{aligned}
$$

Special interchange:

$$
\begin{aligned}
J(f \otimes g) \mathbin{;^{\mathrm{K}}} (m \parallel n) &= (f \otimes g) \mathbin{;} (m \otimes n) \mathbin{;} \psi_{Z,W} \\
&= ((f \mathbin{;} m) \otimes (g \mathbin{;} n)) \mathbin{;} \psi_{Z,W} \\
&= (Jf \mathbin{;^{\mathrm{K}}} m) \parallel (Jg \mathbin{;^{\mathrm{K}}} n)
\end{aligned}
$$

$$
\begin{aligned}
(k \parallel \ell) \mathbin{;^{\mathrm{K}}} J(f \otimes g) &= (k \otimes \ell) \mathbin{;} \psi_{Y,V} \mathbin{;} T(f \otimes g) \\
&= (k \otimes \ell) \mathbin{;} (Tf \otimes Tg) \mathbin{;} \psi_{Z,W} \\
&= ((k \mathbin{;} Tf) \otimes (\ell \mathbin{;} Tg)) \mathbin{;} \psi_{Z,W} \\
&= (k \mathbin{;^{\mathrm{K}}} Jf) \parallel (\ell \mathbin{;^{\mathrm{K}}} Jg)
\end{aligned}
$$

J is an ordered functor:

$$
f \leq g \quad \Rightarrow \quad Jf = f \mathbin{;} \eta_Y \leq g \mathbin{;} \eta_Y = Jg
$$

$$J\mathsf{id}_X = \mathsf{id}_X \,;\, \eta_X = \eta_X = \mathsf{id}^{\mathrm{K}}_X$$

$$J(f \,;\, g) = f \,;\, g \,;\, \eta_Z = f \,;\, \eta_Y \,;\, Tg = f \,;\, \eta_Y \,;\, Tg \,;\, T\eta_Z \,;\, \mu_Z = f \,;\, \eta_Y \,;\, T(g \,;\, \eta_Z) \,;\, \mu_Z = Jf \,;^{\mathrm{K}}\, Jg$$

J is monoidal-like:

$$J\mathsf{I} = \mathsf{I} \,;\, \eta_I \leq \mathsf{I} \,;\, \psi^0 = \psi^0 = \mathsf{jd}$$

$$J(f \otimes g) = (f \otimes g) \,;\, \eta_{Y\otimes V} \leq (f \otimes g) \,;\, (\eta_Y \otimes \eta_V) \,;\, \psi_{Y,V} = ((f \,;\, \eta_Y) \otimes (g \,;\, \eta_V)) \,;\, \psi_{Y,V} = Jf \parallel Jg$$

K is an ordered functor:

$$k \leq \ell \quad \Rightarrow \quad Kk = Tk \,;\, \mu_Y \leq T\ell \,;\, \mu_Y = K\ell$$

$$K\mathsf{id}^{\mathrm{K}}_X = T\eta_X ; \mu_X = \mathsf{id}_{TX}$$

$$K(k ;^{\mathrm{K}} \ell) = T(k ; T\ell ; \mu_Z) ; \mu_Z = TK ; T(T\ell) ; T\mu_Z ; \mu_Z = TK ; T(T\ell) ; \mu_{TZ} ; \mu_Z = Tk ; \mu_Y ; T\ell ; \mu_Z = Kk ; K\ell$$

K is monoidal-like:

$$\psi^0 \,;\, K\mathsf{jd} = \psi^0 \,;\, T\psi^0 \,;\, \mu_I \leq \psi^0 = \mathsf{I} \,;\, \psi^0$$

$$\begin{aligned}
\psi_{X,U} \,;\, K(k \parallel \ell) &= \psi_{X,U} \,;\, T((k \otimes \ell) \,;\, \psi_{Y,V}) \,;\, \mu_{Y\times V} \\
&= \psi_{X,U} \,;\, T(k \otimes \ell) \,;\, T\psi_{Y,V} \,;\, \mu_{Y\times V} \\
&= (Tk \otimes T\ell) \,;\, \psi_{TY,TV} \,;\, T\psi_{Y,V} \,;\, \mu_{Y\times V} \\
&\leq (Tk \otimes T\ell) \,;\, (\mu_Y \otimes \mu_V) \,;\, \psi_{Y,V} \\
&= ((Tk \,;\, \mu_Y) \otimes (T\ell \,;\, \mu_V)) \,;\, \psi_{Y,V} \\
&= (Kk \otimes K\ell) \,;\, \psi_{Y,V}
\end{aligned}$$

$T = K \cdot J$:

$$TX = KX = K(JX)$$

$$Tf = Tf \,;\, T\eta_Y \,;\, \mu_Y = T(f \,;\, \eta_Y) \,;\, \mu_Y = K(Jf)$$

$\varepsilon_X : TX \to X$ in $\mathbf{Kl}(T)$ is natural in X:

$$\varepsilon_X \,;^{\mathrm{K}}\, k = \mathsf{id}_{TX} \,;\, Tk \,;\, \mu_Y = Tk \,;\, T\mathsf{id}_{TY} \,;\, \mu_Y = Tk \,;^{\mathrm{K}}\, \varepsilon_Y$$

J left adjoint to K:

$$\eta_{KX} \,;\, K\varepsilon_X = \eta_{TX} \,;\, T\mathsf{id}_{TX} \,;\, \mu_X = \eta_{TX} \,;\, \mu_X = \mathsf{id}_{TX} = \mathsf{id}_{KX}$$

$$J\eta_X \,;^{\mathrm{K}}\, \varepsilon_{JX} = (\eta_X \,;\, \eta_{TX}) \,;^{\mathrm{K}}\, \mathsf{id}_{TX} = \eta_X \,;\, \eta_{TX} \,;\, T\mathsf{id}_{TX} \,;\, \mu_X = \eta_X \,;\, \eta_{TX} \,;\, \mu_X = \eta_X = \mathsf{id}^{\mathrm{K}}_{JX}$$

□

Proposition 2. *Given an ordered category a $\mathbb{K}$ with an ordered functor $J : \mathbb{C} \to \mathbb{K}$ with an ordered right adjoint K. The ordered functor $T = K \cdot J$ is an ordered monad with unit the unit of the adjunction and multiplication $\mu_X = K\varepsilon_{JX}$.*

Proof. T a monad:

$$\begin{aligned}
\eta_{TX} \,;\, \mu_X &= \eta_{KJX} \,;\, K\varepsilon_{JX} \\
&= \mathsf{id}_{KJX} \\
&= \mathsf{id}_{TX}
\end{aligned}$$

$$\begin{aligned}
\mathsf{id}_{TX} &= \mathsf{id}_{KJX} \\
&= K\mathsf{id}^{\mathrm{K}}_{JX} \\
&= K(J\eta_X \,;^{\mathrm{K}}\, \varepsilon_{JX}) \\
&= KJ\eta_X \,;\, K\varepsilon_{JX} \\
&= T\eta_X \,;\, \mu_X
\end{aligned}$$

$$\begin{aligned}
\mu_{TX} \,;\, \mu_X &= K\varepsilon_{JKJX} \,;\, K\varepsilon_{JX} \\
&= K(\varepsilon_{JKJX} \,;^{\mathrm{K}}\, \varepsilon_{JX}) \\
&= K(JK\varepsilon_{JX} \,;^{\mathrm{K}}\, \varepsilon_{JX}) \\
&= KJK\varepsilon_{JX} \,;\, K\varepsilon_{JX} \\
&= T\mu_X \,;\, \mu_X
\end{aligned}$$

□

Proposition 3. *In the above situation, if J is identity-on-objects, then $\mathbb{K} \cong \mathbf{Kl}(T)$ in $\mathbb{C} \setminus \mathbf{Poset\text{-}Cat}$ (i.e., as ordered categories equipped with ordered functors from $\mathbb{C}$).*

It also suffices that J is bijective-on-objects. And for equivalence instead of isomorphism it suffices that J is essentially surjective.

Proof. It is immediate that $|\mathbb{K}| = |\mathbb{C}| = \mathbf{Kl}(T)$.

We also have that $\mathbb{K}(X,Y) = \mathbb{K}(JX,Y)$ is in bijection with $\mathbb{C}(X,KY) = \mathbb{C}(X,TY) = \mathbf{Kl}(T)(X,Y)$ via the right transpose $(-)^\sharp$ defined by $k^\sharp = \eta_X \mathbin{;} Kk$. The inverse is the left transpose $(-)^\flat$ defined by $k^\flat = Jk \mathbin{;^{\mathrm{K}}} \varepsilon_Y$. Both functions are monotone.

It remains to show that $(-)^\sharp$ gives an ordered isomorphism (in the sense of an ordered functor with an ordered inverse) between the two ordered categories.

$(-)^\sharp$ sends $\mathsf{id}^{\mathrm{K}}_X$ to the Kleisli identity on X of $\mathbf{Kl}(T)$, i.e., η_X:

$$(\mathsf{id}^{\mathrm{K}}_X)^\sharp = \eta_X \mathbin{;} K\mathsf{id}^{\mathrm{K}}_X = \eta_X \mathbin{;} \mathsf{id}_X = \eta_X$$

It sends $k \mathbin{;^{\mathrm{K}}} \ell$ to the Kleisli composition of the images of k and ℓ:

$$\begin{aligned}(k \mathbin{;^{\mathrm{K}}} \ell)^\sharp = \eta_X \mathbin{;} K(k \mathbin{;^{\mathrm{K}}} \ell) &= \eta_X \mathbin{;} Kk \mathbin{;} K\ell \\ &= k^\sharp \mathbin{;} K(J\eta_Y \mathbin{;^{\mathrm{K}}} \varepsilon_Y \mathbin{;^{\mathrm{K}}} \ell) \\ &= k^\sharp \mathbin{;} K(J\eta_Y \mathbin{;^{\mathrm{K}}} JK\ell \mathbin{;^{\mathrm{K}}} \varepsilon_Y) \\ &= k^\sharp \mathbin{;} KJ(\eta_Y \mathbin{;} K\ell) \mathbin{;} K\varepsilon_Y \\ &= k^\sharp \mathbin{;} T\ell^\sharp \mathbin{;} \mu_Y\end{aligned}$$

It also sends Jf to the Kleisli inclusion of f, i.e., $f \mathbin{;} \eta_Y$:

$$(Jf)^\sharp = \eta_X \mathbin{;} KJf = f \mathbin{;} \eta_Y$$

For the next proposition, we also record that $f \mathbin{;} k^\sharp = (Jf \mathbin{;^{\mathrm{K}}} k)^\sharp$ and $(k \mathbin{;^{\mathrm{K}}} Jf)^\sharp = k^\sharp \mathbin{;} Tf$:

$$f \mathbin{;} k^\sharp = f \mathbin{;} \eta_Y \mathbin{;} Kk = \eta_X \mathbin{;} KJf \mathbin{;} Kk = \eta_X \mathbin{;} K(Jf \mathbin{;^{\mathrm{K}}} k) = (Jf \mathbin{;^{\mathrm{K}}} k)^\sharp$$

$$(k \mathbin{;^{\mathrm{K}}} Jf)^\sharp = \eta_X \mathbin{;} K(k \mathbin{;^{\mathrm{K}}} Jf) = \eta_X \mathbin{;} Kk \mathbin{;} KJf = k^\sharp \mathbin{;} Tf$$

□

Proposition 4. *Given a concurrent category $(\mathbb{K}, J)$ (so that in particular J is identity-on-objects) such that J has an ordered right adjoint K. The monad $T = K \cdot J$ is a concurrent monad and $\mathbb{K} \cong \mathbf{Kl}(T)$ as concurrent categories.*

Proof. Define

$$\psi^0 = \mathsf{jd}^\sharp \qquad \psi_{X,Y} = (\varepsilon_X \parallel \varepsilon_Y)^\sharp$$

To establish that $(-)^\sharp$ is an ordered isomorphism to $\mathbf{Kl}(T)$, but also for the proofs below, we need that $\mathsf{jd}^\sharp = \psi^0$ and $(k \parallel \ell)^\sharp = (k^\sharp \otimes \ell^\sharp) \mathbin{;} \psi$. The former is immediate. The latter is proved using special interchange:

$$\begin{aligned}(k \parallel \ell)^\sharp &= ((Jk^\sharp \mathbin{;^{\mathrm{K}}} \varepsilon_Y) \parallel (J\ell^\sharp \mathbin{;} \varepsilon_V))^\sharp \\ &= (J(k^\sharp \otimes \ell^\sharp) \mathbin{;^{\mathrm{K}}} (\varepsilon_Y \parallel \varepsilon_V))^\sharp \\ &= (k^\sharp \otimes \ell^\sharp) \mathbin{;} (\varepsilon_Y \parallel \varepsilon_V)^\sharp \\ &= (k^\sharp \otimes \ell^\sharp) \mathbin{;} \psi_{Y,V}\end{aligned}$$

$\psi_{X,Y}$ is a natural in X, Y, also using special interchange:

$$\begin{aligned}(Tf \otimes Tg) \mathbin{;} \psi_{Y,V} &= (Tf \otimes Tg) \mathbin{;} (\varepsilon_Y \parallel \varepsilon_V)^\sharp \\ &= (J(Tf \otimes Tg) \mathbin{;^{\mathrm{K}}} (\varepsilon_Y \parallel \varepsilon_V))^\sharp \\ &= ((JKJf \mathbin{;^{\mathrm{K}}} \varepsilon_Y) \parallel (JKJg \mathbin{;^{\mathrm{K}}} \varepsilon_V))^\sharp \\ &= ((\varepsilon_X \mathbin{;^{\mathrm{K}}} Jf) \parallel (\varepsilon_Y \mathbin{;^{\mathrm{K}}} Jg))^\sharp \\ &= ((\varepsilon_X \parallel \varepsilon_Y) \mathbin{;^{\mathrm{K}}} J(f \otimes g))^\sharp \\ &= (\varepsilon_X \parallel \varepsilon_Y)^\sharp \mathbin{;} T(f \otimes g) \\ &= \psi_{X,Y} \mathbin{;} T(f \otimes g)\end{aligned}$$

T is a lax monoidal functor, also using special interchange:

$$\begin{array}{rcl}(\psi^0 \otimes \mathsf{id}_{TX})\,;\psi_{\mathsf{I},X}\,;T\lambda_X &=& (\mathsf{jd}^\sharp \otimes \mathsf{id}_{TX})\,;(\varepsilon_{\mathsf{I}} \parallel \varepsilon_X)^\sharp\,;T\lambda_X\\ &=& (\mathsf{jd}^\sharp \otimes \mathsf{id}_{TX})\,;((\varepsilon_{\mathsf{I}} \parallel \varepsilon_X)\,;^{\mathrm{K}}\,J\lambda_X)^\sharp\\ &=& (J(\mathsf{jd}^\sharp \otimes \mathsf{id}_{TX})\,;^{\mathrm{K}}\,(\varepsilon_{\mathsf{I}} \parallel \varepsilon_X)\,;^{\mathrm{K}}\,J\lambda_X)^\sharp\\ &=& (((J\mathsf{jd}^\sharp\,;^{\mathrm{K}}\,\varepsilon_{\mathsf{I}}) \parallel (J\mathsf{id}_{TX}\,;^{\mathrm{K}}\,\varepsilon_X))\,;^{\mathrm{K}}\,J\lambda_X)^\sharp\\ &=& ((\mathsf{jd}^\sharp)^\flat \parallel \varepsilon_X)\,;^{\mathrm{K}}\,J\lambda_X)^\sharp\\ &=& ((\mathsf{jd} \parallel \varepsilon_X)\,;^{\mathrm{K}}\,J\lambda_X)^\sharp\\ &=& (J\lambda_{TX}\,;^{\mathrm{K}}\,\varepsilon_X)^\sharp\\ &=& \lambda_{TX}\,;\varepsilon_X^\sharp\\ &=& \lambda_{TX}\end{array}$$

$$\begin{array}{rcl}T\rho_X &=& \varepsilon_X^\sharp\,;T\rho_X\\ &=& (\varepsilon_X\,;^{\mathrm{K}}\,J\rho_X)^\sharp\\ &=& (J\rho_{TX}\,;^{\mathrm{K}}\,(\varepsilon_X \parallel \mathsf{jd}))^\sharp\\ &=& (J\rho_{TX}\,;^{\mathrm{K}}\,(\varepsilon_X \parallel (\mathsf{jd}^\sharp)^\flat))^\sharp\\ &=& (J\rho_{TX}\,;^{\mathrm{K}}\,((J\mathsf{id}_{TX}\,;^{\mathrm{K}}\,\varepsilon_X) \parallel (J\mathsf{jd}^\sharp\,;^{\mathrm{K}}\,\varepsilon_{\mathsf{I}})))^\sharp\\ &=& (J\rho_{TX}\,;^{\mathrm{K}}\,J(\mathsf{id}_{TX} \otimes \mathsf{jd}^\sharp)\,;^{\mathrm{K}}\,(\varepsilon_X \parallel \varepsilon_{\mathsf{I}}))^\sharp\\ &=& \rho_{TX}\,;(J(\mathsf{id}_{TX} \otimes \mathsf{jd}^\sharp)\,;^{\mathrm{K}}\,(\varepsilon_X \parallel \varepsilon_{\mathsf{I}}))^\sharp\\ &=& \rho_{TX}\,;(\mathsf{id}_{TX} \otimes \mathsf{jd}^\sharp)\,;(\varepsilon_X \parallel \varepsilon_{\mathsf{I}})^\sharp\\ &=& \rho_{TX}\,;(\mathsf{id}_{TX} \otimes \psi^0)\,;\psi_{X,\mathsf{I}}\end{array}$$

$$\begin{array}{rcl}(\psi_{X,Y} \otimes \mathsf{id}_{TZ})\,;\psi_{X\otimes Y,Z}\,;T\alpha_{X,Y,Z} &=& ((\varepsilon_X \parallel \varepsilon_Y)^\sharp \otimes \mathsf{id}_{TZ})\,;(\varepsilon_{X\otimes Y} \parallel \varepsilon_Z)^\sharp\,;T\alpha_{X,Y,Z}\\ &=& ((\varepsilon_X \parallel \varepsilon_Y)^\sharp \otimes \mathsf{id}_{TZ})\,;((\varepsilon_{X\otimes Y} \parallel \varepsilon_Z)\,;^{\mathrm{K}}\,J\alpha_{X,Y,Z})^\sharp\\ &=& (J((\varepsilon_X \parallel \varepsilon_Y)^\sharp \otimes \mathsf{id}_{TZ})\,;^{\mathrm{K}}\,(\varepsilon_{X\otimes Y} \parallel \varepsilon_Z)\,;^{\mathrm{K}}\,J\alpha_{X,Y,Z})^\sharp\\ &=& (((J(\varepsilon_X \parallel \varepsilon_Y)^\sharp\,;^{\mathrm{K}}\,\varepsilon_{X\otimes Y}) \parallel (J\mathsf{id}_{TZ}\,;^{\mathrm{K}}\,\varepsilon_Z))\,;^{\mathrm{K}}\,J\alpha_{X,Y,Z})^\sharp\\ &=& ((((\varepsilon_X \parallel \varepsilon_Y)^\sharp)^\flat \parallel \varepsilon_Z)\,;^{\mathrm{K}}\,J\alpha_{X,Y,Z})^\sharp\\ &=& (((\varepsilon_X \parallel \varepsilon_Y) \parallel \varepsilon_Z)\,;^{\mathrm{K}}\,J\alpha_{X,Y,Z})^\sharp\\ &=& (J\alpha_{TX,TY,TZ}\,;^{\mathrm{K}}\,(\varepsilon_X \parallel (\varepsilon_Y \parallel \varepsilon_Z)))^\sharp\\ &=& (J\alpha_{TX,TY,TZ}\,;^{\mathrm{K}}\,((\varepsilon_X \parallel (\varepsilon_Y \parallel \varepsilon_Z)^\sharp)^\flat))^\sharp\\ &=& (J\alpha_{TX,TY,TZ}\,;^{\mathrm{K}}\,((J\mathsf{id}_{TX}\,;^{\mathrm{K}}\,\varepsilon_X) \parallel (J(\varepsilon_Y \parallel \varepsilon_Z)^\sharp\,;^{\mathrm{K}}\,\varepsilon_{Y\otimes Z}))^\sharp\\ &=& (J\alpha_{TX,TY,TZ}\,;^{\mathrm{K}}\,J(\mathsf{id}_{TX} \otimes (\varepsilon_Y \parallel \varepsilon_Z)^\sharp)\,;^{\mathrm{K}}\,(\varepsilon_X \parallel \varepsilon_{Y\otimes Z}))^\sharp\\ &=& \alpha_{TX,TY,TZ}\,;(J(\mathsf{id}_{TX} \otimes (\varepsilon_Y \parallel \varepsilon_Z)^\sharp)\,;^{\mathrm{K}}\,(\varepsilon_X \parallel \varepsilon_{Y\otimes Z}))^\sharp\\ &=& \alpha_{TX,TY,TZ}\,;(\mathsf{id}_{TX} \otimes (\varepsilon_Y \parallel \varepsilon_Z)^\sharp)\,;(\varepsilon_X \parallel \varepsilon_{Y\otimes Z})^\sharp\\ &=& \alpha_{TX,TY,TZ}\,;(\mathsf{id}_{TX} \otimes \psi_{Y,Z})\,;\psi_{X,Y\otimes Z}\end{array}$$

T satisfies interchanges:

$$\begin{array}{rcl}\eta_I &=& \mathsf{id}_I^{\mathrm{K}\sharp}\\ &\leq& \mathsf{jd}^\sharp\\ &=& \psi^0\end{array}$$

$$\begin{array}{rcl}\psi^0\,;T\psi^0\,;\mu_{\mathsf{I}} &=& \mathsf{jd}^\sharp\,;T\mathsf{jd}^\sharp\,;\mu_I\\ &=& (\mathsf{jd}\,;^{\mathrm{K}}\,\mathsf{jd})^\sharp\\ &\leq& \mathsf{jd}^\sharp\\ &=& \psi^0\end{array}$$

$$\eta_{X\otimes Y} \;=\; \mathsf{id}_{X\otimes Y}^{\mathrm{K}\,\sharp}$$

$$
\begin{array}{rcl}
& \leq & (\mathsf{id}_X^{\mathrm{K}} \parallel \mathsf{id}_Y^{\mathrm{K}})^\sharp \\
& = & (\mathsf{id}_X^{\mathrm{K}\sharp} \otimes \mathsf{id}_Y^{\mathrm{K}\sharp}) \,;\, \psi_{X,Y} \\
& = & (\eta_X \otimes \eta_Y) \,;\, \psi_{X,Y}
\end{array}
$$

$$
\begin{array}{rcl}
\psi_{TX,TY} \,;\, T\psi_{X,Y} \,;\, \mu_{X\otimes Y} & = & (\varepsilon_{TX} \parallel \varepsilon_{TY})^\sharp \,;\, T(\varepsilon_X \parallel \varepsilon_Y)^\sharp \,;\, \mu_{X\otimes Y} \\
& = & ((\varepsilon_{TX} \parallel \varepsilon_{TY}) \,;^{\mathrm{K}}\, (\varepsilon_X \parallel \varepsilon_Y))^\sharp \\
& \leq & ((\varepsilon_{TX} \,;^{\mathrm{K}}\, \varepsilon_X) \parallel (\varepsilon_{TY} \,;^{\mathrm{K}}\, \varepsilon_Y))^\sharp \\
& = & ((JK\varepsilon_X \,;^{\mathrm{K}}\, \varepsilon_X) \parallel (JK\varepsilon_Y \,;^{\mathrm{K}}\, \varepsilon_Y))^\sharp \\
& = & (J(K\varepsilon_X \otimes K\varepsilon_Y) \,;^{\mathrm{K}}\, (\varepsilon_X \parallel \varepsilon_Y))^\sharp \\
& = & (K\varepsilon_X \otimes K\varepsilon_Y) \,;\, (\varepsilon_X \parallel \varepsilon_Y)^\sharp \\
& = & (\mu_X \otimes \mu_Y) \,;\, \psi_{X,Y}
\end{array}
$$

□